Data Dictionary — Schema Specification

HCLTech Career Shaper Programme | Project 81: Agentic Web Scraper and Market Intelligence Reporter

Target File: dataset.csv	**Total Columns:** 7	**Null Constraint:** 0 Nullable Columns (100% Complete)
Domain: Agentic AI / Python	**Total Benchmark Records:** 5,000	**Status Values:** success, partial, failed

Column Name	Data Type	Sample 1	Sample 2	Sample 3	Null?	Field Description & Business Rules
`run_id`	String (Alphanumeric)	RUN0001	RUN0002	RUN0003	No	Unique identifier assigned to each automated web scraping execution run; format RUN followed by 4 digits.
`target_domain`	String (FQDN)	hcltech.com	tcs.com	infosys.com	No	Fully qualified domain name of the web target crawled and analyzed by the scraping agent across 10 industry sectors (e.g., hcltech.com, tcs.com, bloomberg.com).
`pages_crawled`	Integer (Count)	432	403	13	No	Total number of distinct HTML web pages discovered, fetched, and traversed during the session.
`data_extracted_rows`	Integer (Count)	5186	9556	2511	No	Total count of structured market intelligence data records successfully extracted from crawled web pages.
`scrape_duration_s`	Float (Seconds)	397.0	526.2	343.0	No	Total wall-clock duration of the web crawl, HTML cleaning, and entity extraction pipeline in seconds.
`status`	String (Enum: success | partial | failed)	partial	success	success	No	Operational execution outcome of the scrape job: 'success' (completed), 'partial' (degraded/rate-limited), or 'failed'.
`report_generated`	Integer / Boolean (0 or 1)	1	0	1	No	Binary indicator flag where 1 indicates an AI market intelligence report was generated and 0 indicates report omitted.

Learner Implementation Notes & Schema Constraints:

1. **Primary Key:** run_id is guaranteed unique across all executions.
2. **Referential Integrity:** target_domain represents target websites crawling under strict robots.txt compliance.
3. **Data Quality Verification:** Extracted rows must be ≥ 0. Zero values on failed status represent timeout or block occurrences.
4. **Boolean Representation:** report_generated uses strict binary encoding (1 = true, 0 = false) for seamless statistical aggregation in Pandas.